Multivitamin Compliance Reduces Injuries of Female Recruits at Air Force Basic Training: A Randomized Controlled Cohort Study

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ABSTRACT

Introduction:

Department of the Air Force basic military training (BMT) is the 7.5-week initial entry training for all enlisting U.S. Air Force and Space Force recruits. Overuse musculoskeletal injuries (MSKI) during training threaten success in BMT, and trainees with nutritional deficiencies are at higher risk. Several efforts are made at BMT to mitigate these issues to help trainees graduate on time, such as the distribution of multivitamins (MVI) to female recruits. However, trainee compliance with the prescribed MVI has been reported to be low, calling into question the effectiveness of this intervention. This study evaluated the effect of briefing modality offered to these trainees at the time of distribution on educational effectiveness, MVI compliance, and injury rates.

Materials and Methods:

This was a randomized controlled cohort study of BMT female trainees. Subjects included all female trainees of a single iteration of BMT concurrently completing training within the same squadron. At the outset of training, cohorts were randomly educated in one of two ways regarding the provided MVI. The control cohort ($n = 80$) was presented with the standard educational video created by BMT-registered dietitians, while the test cohort ($n = 79$) was presented the video and then also briefed in-person by a group of subject matter experts. At the end of BMT, the trainees completed a questionnaire regarding their knowledge, confidence, and compliance with the MVI. Additionally, a routinely collected training database was queried and analyzed for differences in injuries and training outcomes using the chi-squared test with significance set to $\alpha \leq 0.05$.

Results:

Although both cohorts found their educational presentation useful, the "in-person" cohort reported significantly higher compliance rates and were found to be diagnosed with MSKI and bone stress injuries at significantly lower rates (43% and 66% relative risk reductions, respectively).

Conclusions:

Education to female BMT trainees regarding the MVI, which included a face-to-face discussion with subject matter experts, resulted in greater compliance and reduced incidence of MSKI and bone stress injury, compared to a video prepared by registered dietitians alone. Such educational approaches should continue to be strived for in this and other military population health measures. This study's implication of MVI efficacy for injury risk reduction in female trainees should be further studied, verified, and improved upon in this and other populations.

INTRODUCTION

Nutritional deficiency is common among military trainees[1–5]at the Department of the Air Force basic military training (BMT), the 7.5-week physically, mentally, and emotionally demanding initial training for recruits seeking enlistment into the U.S. Air Force and U.S. Space Force. Deficiencies occurring before BMT (baseline nutrition) and during BMT (relative to training demands or in response to exercise and adaptations) can interfere with training success via effects on musculoskeletal injury (MSKI),[6] illness,[7–9] and poor mental and physical adaptation and performance.[4,10–19]

Bone stress injury (BSI) is a specific overuse MSKI very common in military recruits, to which those harboring a nutritional deficiency are at higher risk.[20–23] In a recent publication, 13.87% of male trainees and 14.13% of female trainees were clinically diagnosed with BSI at BMT.[24] These conditions result in increased health care costs, training disruptions, delayed graduation ($384.33/day/trainee), and incompletion of training rates (~$55 K/trainee attrition).[25] BSI and

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its nutritional risk factors (such as low-energy availability,[26] iron,[5,22,27] and calcium/vitamin D[28,29]) thus present important targets for prevention and mitigation efforts in military training.

BSI-facilitating nutritional deficits in BMT trainees occur for myriad reasons which have been reported clinically by trainees to the clinicians of this authorship team. Before BMT, trainees may be affected by food insecurity, knowledge deficits, and restricted diets. During BMT, intake of adequate nutrition may be hindered by unavailability of preferred or religious food choices at the dining facilities (DFAC), lack of adequate mealtime in the DFAC, and appetite reductions associated with the stress of training. Regardless of the reasons, trainees with intentional or unintentional nutritional deficits at BMT are predisposed to worse training adaptation, which may lead to hindered performance as well as MSKI; such trainees are at higher risk to not graduate BMT on time or at all.[30]

For these reasons, BMT has taken steps to improve the fueling of trainees. DFAC food product offerings have improved to better meet nutritional needs and specific diet requirements of individuals. Furthermore, in BMT DFACs, changes to choice architecture and the introduction of the Go For Green® program,[31] which provides quick assessments of menu item nutritional value, have helped direct trainees to healthier choices. Beyond their three daily meals in the DFAC, since 2012, trainees have been provided an evening snack in the form of their choice of a fruit or nutrition bar "to improve nutritional adequacy and timing in relation to exercise and as well as reduce attrition."[4]

Female military trainees are at higher risk of nutritional deficiency, most notably iron deficiency,[5,32–34] injury,[23,30] and attrition[35] compared to their male counterparts. To address these disproportionate risks and "improve performance and decrease rates of attrition," a daily prenatal multivitamin (MVI) with iron has been prescribed and distributed to females arriving to BMT, since 2012.[35] Of note, this MVI is labeled "prenatal" because of its folic acid content (which is recommended for all females of child-bearing potential per the Centers for Disease Control and Prevention[36] and the United States Preventive Services Task Force[37]) and was chosen because prenatal MVIs are on the organization's formulary whereas other MVIs are not. However, the MVI was chosen specifically for vitamin D and iron supplementation. The contents of this MVI are listed in Table I. Though the vitamin D content (400 IU) is half the dose associated with decreased stress fractures among navy recruits reported by Lappe et al.,[28] the 27 mg of iron approximates the estimated 22 mg daily requirement for females in military training.[8]

A study by Barnes et al.[35] retrospectively investigated the effects of prescribing this MVI to female trainees and "found a statistically nonsignificant effect of prenatal vitamin supplementation on attrition from training and on development of stress fractures and iron deficiency anemia among females undergoing U.S. Air Force BMT" but deemed the 26% associated reduction in medical attrition as "operationally adequate to justify continuation" of the program. Those authors noted limitations from its nonrandomized, retrospective nature comparing trainees who were prescribed the prenatal and counseled on its potential benefits and risks with a historical season-matched control group that previously matriculated BMT and lack of assessment of compliance. A 2014 survey of female BMT trainees found that less than half reported daily MVI compliance.[35]

TABLE I. Supplement Facts of the Prenatal MVI Distributed to Females at BMT

Vitamin or mineral	Amount per serving (1 tablet), taken once daily	Percent daily value for pregnant and lactating women (%)
Vitamin A (as acetate and 4% beta carotene)	1,200 μg	92
Vitamin C (as ascorbic acid)	100 μg	83
Vitamin D (as cholecalciferol)	10 μg (400 IU)	67
Vitamin E (as alpha tocopherol acetate)	5 mg	26
Thiamin (as thiamine mononitrate)	1.5 mg	107
Riboflavin	1.7 mg	106
Niacin (as niacinamide)	18 mg	100
Vitamin B6 (as pyridoxine HCl)	2.6 mg	130
Folate	1,360 μg(800 μg folic acid)	227
Vitamin B12 (as cyanocobalamin)	4 μg	143
Calcium (as calcium carbonate and calcium sulfate)	200 mg	15
Iron (as ferrous fumarate)	27 mg	100
Zinc (as zinc oxide)	25 mg	192

Despite the statistically nonsignificant findings, MVI prescriptions for female trainees have continued at BMT with briefings given in various forms. The process of education and distribution of this MVI occurs during medical in-processing, during the first week of BMT; since 2020, groups of female trainees have been shown a video put together by embedded registered dietitians (RD). In the video, BMT RDs review the purpose of the MVI, its contents, potential benefits, and how and when to take it. Trainees may ask questions of an enlisted medical technician before the distribution of a 3-month supply of the MVI. However, anecdotally, many trainees reported poor compliance with the MVI, either by choice or forgetfulness.

Countless questions remain regarding the effectiveness and efficacy of dietary supplementation for injury and performance improvements, and specifically within military trainees for improved operational outcomes. Despite the effort

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and resources appropriated for providing an MVI with iron to all female trainees, the intended benefits have not conclusively been demonstrated.[35] Issues with compliance as well as nutrient timing and dosage in relation to the high stresses of BMT may preclude realization of performance, injury, and attrition benefits. This study aimed to determine whether the addition of an in-person brief of the MVI by subject-matter experts (SMEs) at the time of distribution would improve female BMT trainee knowledge, confidence, and compliance with the MVI and whether this, in turn, would affect MSKI/BSI incidence and training outcomes. On the other hand, improved compliance without a corresponding improvement of injury incidence or training outcomes could encourage BMT trainee health leadership to support discontinuation the MVI process at BMT for cost and time savings.

METHODS

Every recruit arriving to BMT in the same week is assigned to the same "squadron" for housing and training and are further divided into "flights" (groups of ~40 trainees of the same gender). Female trainees medically in-process as a unit with their flight mates, during which time they receive the MVI. The start date for a recruit is essentially random as is the flight to which they are assigned by BMT leadership.

All female trainees from a single training squadron who concurrently entered BMT in December 2021, and who had already been randomly assigned to 1 of the 4 flights, were enrolled in this command-directed process improvement evaluation. These flights were randomly assigned to cohorts, which would receive one of two MVI-briefing modalities. The control cohort consisted of every recruit within 2 female trainee flights, which received standard MVI education via the BMT RD video with medical technician facilitation ("video-only" cohort). The test cohort consisted of every recruit within the other 2 female trainee flights, which were presented the same video and reviewed important information with an in-person team of SMEs (the trainee health clinic medical director/sports medicine physician, RD, and diet technician) and were given an opportunity to ask questions ("in-person" cohort). Because this study was designed as an internal, commander-directed process improvement evaluation, and obtaining individual consent from each member of the squadron was deemed overly burdensome and costly to the training and medical processing schedule, trainees were not made aware that their briefing experience was different from that of any others; this was further deemed crucial since trainee knowledge of their compliance being assessed would likely skew their compliance and report thereof. Flight and squadron leadership were also not made aware of the change in education, nor were other medical providers. No other cohort differences with regards to MVI distribution, education, training, or medical care occurred for the entirety of BMT. After the completion of BMT, injury, performance, and outcome data (regularly tracked for all BMT trainees) were compiled for these female flights; this and the survey results were then compared between the flights that received the video-only or the in-person MVI education. There was no crossing over between flights/cohorts. The Institutional Review Board at the 59th Medical Wing at JBSA-Lackland approved the project as an exempt protocol (FWH20220101E).

TABLE II. Demographic Characteristics of Participants

Demographic	Video-only cohort Mean ± SD	In-person cohort Mean ± SD
Total, *n*	80	79
Age, year	21 ± 3	23 ± 5
BMI	24.1 ± 2.6	24.0 ± 2.8

Values are reported as averages ± standard deviations (SD).
BMI calculated using height (meters) and weight (kilograms).

Multiple routinely collected and manually linked (person level, via full name and DoD identification number) databases were queried for this study. The Electronic Basic Training Management System is maintained by Air Education and Training Command and stores demographic data such as squadron and flight assignment, age, gender (as identified in the Defense Enrollment Eligibility Reporting System), body mass index (BMI), and training outcome data (e.g., attrition and training delays). A database of MSKI diagnosed at BMT is maintained by sports medicine staff, which is routinely analyzed by members of the study team for injury trend monitoring and program improvement. Furthermore, all trainees completed questionnaires regarding their perceptions regarding the usefulness and effectiveness of their MVI briefing and were asked to quantify their MVI compliance. All data linkage and analysis (injury, training delays, graduation, briefing perceptions, and compliance) occurred following completion of the training period.

For this prospective, randomized controlled cohort study, a power analysis calculation was completed using $\alpha \leq 0.05$ for the significance level, a power of 0.99. We estimated a 25% relatively greater daily MVI compliance in trainees receiving the in-person brief compared to those viewing the video alone, which yielded a need for each cohort to have 77 trainees to demonstrate a significant difference.

Associations between categorical variables were analyzed using the chi-squared test. Significance was set to $\alpha \leq 0.05$. Statistical analyses were performed using SAS version 9.4 (Statistical Analysis Software, Cary, NC).

RESULTS

Demographic characteristics of the 159 female trainees (80 in the video-only cohort, and 79 in the in-person cohort) subjected to this study are provided in Table II. No significant differences in age or BMI between cohorts were detected.

Results from the survey regarding briefing effectiveness and MVI compliance as well as routinely collected data regarding injuries and training delays of the participants are

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TABLE III. Survey, Injury, and Training Results

	Video-only cohort ($n = 80$) (%)	In-person cohort ($n = 79$) (%)	*P*
"Brief Effectiveness (survey)"			
Found brief effective in explaining why MVI prescribed	95.89	98.63	.31
Confident in understanding why MVI was prescribed	90.41	100	**.007**
"Compliance (survey)"			
Took MVI daily	14.86	34.25	**.0063**
Took MVI most days	37.84	63.01	**.0023**
Never took MVI	20.27	8.22	**.037**
"Injuries (routinely collected data)"			
MSKI	31.25	17.72	**.047**
BSI	15	5.06	**.037**
"Training Delays (routinely collected data)"			
Medical hold	1.27	1.27	.99
Medical hold: MSKI	0	0	–
Recycle: admin issue	0	1.27	.31
Recycle: fitness failure	2.50	2.53	.99

The bold values represent statistical significance ($\alpha \leq 0.05$).

summarized in Table III. There were no losses or exclusions of routinely collected and linked data.

In regard to the briefing, there were no significant differences between cohorts in proportion of trainees reporting that they found the brief to be effective in explaining why the MVI was prescribed. However, there was a significantly larger proportion of those within the cohort that received the in-person brief who stated they felt confident in their understanding of why they were prescribed the MVI (100% vs. 90.41%, $P = .007$).

In regard to compliance, compared to the video-only cohort, a significantly higher proportion within the in-person cohort, a significantly higher proportion within the in-person cohort reported taking the MVI on most days (or more) (63.01% vs. 37.84%, $P = .0023$), and a significantly lower proportion reported never taking the MVI (8.22% vs. 20.27%, $P = .037$).

In regard to injury, a significantly smaller proportion of the in-person cohort suffered MSKI (17.72% vs. 31.25%, $P = .047$) and BSI (5.06% vs. 15%, $P = .037$) compared to the video-only cohort.

In regard to training delays, there were no significant differences of proportion of trainees who were sent to medical hold or recycled in training between cohorts.

DISCUSSION

From the results of this study, we infer that incorporating an in-person brief from SMEs improved compliance with the MVI which in turn significantly reduced injuries. The relative and absolute risk reductions for diagnosis with all MSKI were 43% and 13.5%, respectively. This absolute risk reduction corresponds to a number needed to in-person brief (NNIPB) of 7.4 to avoid one MSKI. Said another way, with ∼40 female trainees in a briefing, each in-person brief delivered would be expected to prevent 5.4 MSKI compared to the video-only brief. The relative and absolute risk reductions for diagnosis with BSI were 66% and 10%, respectively; this corresponds to a NNIPB of 10. Said another way, each in-person brief (of 40 female trainees) would be expected to prevent four BSIs. Based on current literature on nutritional deficiency, dietary supplementation, and BSI risk, the iron[22,33] and vitamin D/calcium[28] content of the MVI is suspected to have been most beneficial to BSI reduction in this study. Though less is known regarding the association with injury and efficacy for risk reduction of other vitamins and minerals in the MVI, they may also have supported healthy tissue adaptations to training stress.

Although neither this prospective study nor the retrospective study by Barnes et al. detected significant differences in training disruption, we showed a significant difference in BSI, whereas Barnes et al. did not.[35] This is likely due to this study's randomized nature as opposed to using a historical, season-matched comparison group as the control. Our design controlled for temporal fluctuations in diagnostic patterns, access to care, care-seeking behaviors, and training activities and policies, which may have skewed the findings of the prior study. The lack of significant difference in attrition despite differences in BSI, as was also the finding in a recent randomized controlled trial by Kasper et al.,[24] is a testament to the early identification and efficient rehabilitation of MSKI provided by the BMT sports medicine team, which was implemented specifically to mitigate the operational impact of injury at BMT.[25]

This study was limited in several ways: (1) it used block rather than individual randomization, (2) all trainees received the MVI (because of the potential benefits of the MVI), (3) it did not collect data regarding existing nutritional deficiencies, and (4) it was only conducted on a single BMT iteration so it cannot be known how the difference in season may affect the outcome. Despite these limitations, it adds to the body of knowledge regarding the benefits of this approach in female military trainees. Besides the presence and engagement of the SMEs at the briefing, to the authors' knowledge, no other differences in training or healthcare occurred between these two groups to explain the decrease in BSI. Flights within a squadron undergo nearly identical training, tissue loads, and access to medical care (the same embedded athletic trainers and sports medicine physicians), so it is unlikely that being in different flights alone would account for the differences in injury diagnoses. A 2022 study of the implementation of mass running gait training at BMT found that trainees who experienced a face-to-face encounter with medical personnel, who introduced themselves as available to help with issues related to BMT completion, reduced hesitancy to be seen, and subsequently increased BSI diagnoses.[38] With a similar interaction between trainees and embedded medical staff in our study potentially also increasing trainee willingness

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to report symptoms, resultant injury diagnoses could have been promoted among the in-person MVI-briefing cohort,[38] but there were still significantly fewer injury diagnoses in this cohort. Controlling for this possible confounder in future work may reveal a larger decrease in injuries attributable to MVI compliance alone.

Overweight and obesity is common in the U.S. Military (following training), and programs are in place throughout the DoD to address this.[39] However, it is important to remember the nutritionally related health and performance issues related to the underconsumption of nutrients within military training situations to fuel necessary performance and required physical adaptations[17]; nutritional programs must be specific to the particular setting or goal (training vs. readiness maintenance). Interventions aimed at overcoming deficiencies relative to training and providing the timely delivery of relevant health and performance nutrition education and interventions are vital in improving success rates at BMT. Nutritional supplementation in military training should be further studied to verify its efficacy in injury reduction in at-risk populations and improve formulations specific to the needs of the population.

At the time of this study, the standard was to brief trainees on the MVI via video, which avoided repetitively burdening the clinical staff with being present for multiple briefings each week at various times. Although simplification and automation are often valuable for repetitive processes, this study shows that an automated (video-only) MVI briefing at BMT was associated with worse compliance and injury rates than an in-person briefing. In-person medical intervention education by interested SMEs, albeit more burdensome, has demonstrable benefits on the learner's confidence in their understanding and compliance. The cost and benefits of educational modalities regarding individual or population health measures should be carefully considered, and whenever feasible and cost effective, face-to-face educational approaches should be undertaken. Since this study, embedded RDs have regularly delivered the BMT MVI briefings.

This study is but a glimpse of the benefits to military training that are possible with improved embedded nutrition services. Future research should evaluate the benefit of embedded nutrition assets at BMT on training outcomes. Fisher et al.[25] showed that embedding sports medicine services at BMT improved on-time graduation rates and reduced costs, time out of training, and attrition. Embedded nutritional medicine assets at BMT may be similarly beneficial by providing sports performance nutrition education, reducing barriers to nutrition care, and working toward improved DFAC offerings and environments. The efficient production of fit and healthy airmen at BMT prepared for military service would be best served by continuing to assess and improve trainee performance health education regarding balanced, healthy diets that meet the nutritional needs of their diverse lifestyles and evolving careers.

CONCLUSION

This study shows that including face-to-face education to female BMT trainees regarding a MVI by SMEs resulted in greater compliance and reduced MSKI and BSI incidence, compared to a video prepared by RDs alone. Such educational approaches should continue to be strived for in this and other military population health measures. This study's implication of MVI efficacy for injury risk reduction in female military trainees should be further studied, verified, and improved upon in this and other populations.

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INSTITUTIONAL REVIEW BOARD (HUMAN SUBJECTS)

The IRB of the 59 MDW approved of the exempt status of this project.

INDIVIDUAL AUTHOR CONTRIBUTION STATEMENT

K.B.K. designed this research, analyzed the data, and drafted the original manuscript. M.C. analyzed data, reviewed, and edited the manuscript. C.E. and R.G. collected data, reviewed, and edited the manuscript. All authors read and approved the final manuscript.

INSTITUTIONAL CLEARANCE

59 MDW Clearance approved.

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CONFLICT OF INTEREST STATEMENT

None declared.

DATA AVAILABILITY

All relevant data are included in manuscript.

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